

OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-216 — Kudzu Plus Capsules** (amazon.de). Prepared for review.

1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is $P(0)=e^{-\lambda}$: ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	3 <= reach < 30 days AND throttled (price >= +8% and/or ad spend cut >= 50%)	Miss (voluntary)
4	Marketplace gap	Units = 0 on a day enclosed by sales, lambda >= 3/day	Lost (involuntary)

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = max(lambda - actual units, 0)** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

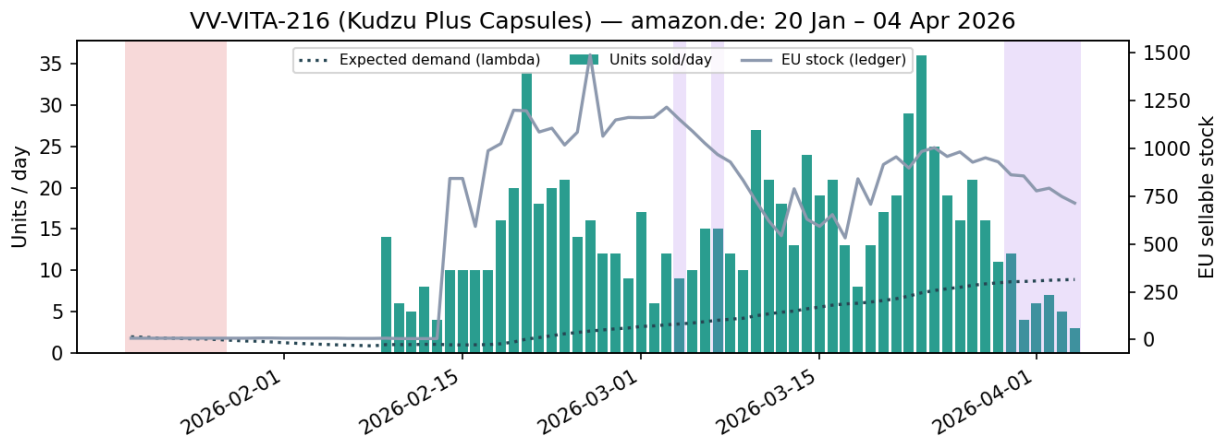
Lost / Miss revenue = missed units x avg price

Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-216 — Kudzu Plus Capsules, amazon.de. Full year on this marketplace: **67 OOS days** (lost €4.948 revenue / €1.572 CM3) and **43 cooling-down days** (miss €6.448 revenue / €1.508 CM3).



Highlighted window 20 Jan 2026 – 04 Apr 2026: EU stock ranged from ~1,490 down to ~5 units and reach bottomed at ~1 day(s). In this window the model flagged **8 OOS day(s)** (red/orange shading) and **8 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
01-20	0	2	7	1	—	Critically low (<3d)	12	
01-21	0	2	7	1	—	Critically low (<3d)	12	
01-22	0	2	7	1	—	Critically low (<3d)	11	
01-23	0	2	7	2	—	Critically low (<3d)	11	
01-24	0	2	7	2	—	Critically low (<3d)	11	
01-25	0	2	7	2	—	Critically low (<3d)	11	
01-26	0	2	7	2	—	Critically low (<3d)	11	
01-27	0	2	7	3	—	Critically low (<3d)	10	
01-28	0	2	7	3	—	—		
02-12	8	1	5	23	19.97	—		
02-27	12	3	1149	40	19.99	—		
03-04	9	3	1152	29	19.90	Cooling down		
03-07	15	4	968	21	20.12	Cooling down		
03-16	21	6	654	9	19.88	—		
03-30	12	9	862	12	19.89	Cooling down		
03-31	4	9	856	12	19.90	Cooling down		22
04-01	6	9	778	11	19.99	Cooling down		13
04-02	7	9	792	11	19.79	Cooling down		8
04-03	5	9	748	11	20.53	Cooling down		18
04-04	3	9	714	10	19.90	Cooling down		27

6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.